
CoSimGen: Controllable diffusion model for simultaneous image and segmentation mask generation

Abstract

1 Generating paired images and segmentation masks remains a core bottleneck in
2 data-scarce domains such as medical imaging and remote sensing, where manual
3 annotation is expensive, expertise-dependent, and ethically constrained. Existing
4 generative approaches typically handle image or mask generation in isolation and
5 offer limited control over spatial and semantic outputs. We introduce CoSimGen,
6 a diffusion-based framework for controllable simultaneous generation of images
7 and segmentation masks. CoSimGen integrates multi-level conditioning via (1)
8 class-grounded textual prompts enabling hot-swapping of input control, (2) spatial
9 embeddings for contextual coherence, and (3) spectral timestep embeddings for
10 denoising control. To enforce alignment and generation fidelity, we combine con-
11 trastive triplet loss between text and class embeddings with diffusion and adversarial
12 objectives. Low-resolution outputs (128×128) are super-resolved to 512×512 ,
13 ensuring high-fidelity synthesis. Evaluated across five diverse datasets, CoSimGen
14 achieves state-of-the-art performance in FID, KID, LPIPS, and Semantic-FID, with
15 KID as low as 0.11 and LPIPS of 0.53. Our method enables scalable, controllable
16 dataset generation and advances multimodal generative modeling in structured
17 prediction tasks.

18 1 Introduction

19 Creating large-scale paired datasets of images and segmentation masks is a major bottleneck in do-
20 mains like medical imaging [1], geospatial analysis [2], autonomous driving [9], and surgical AI [22].
21 Manual annotation is costly, domain-specific, and often ethically constrained. While generative
22 models such as VAEs [18], GANs [11], and diffusion models [32, 14] have advanced image synthesis,
23 most methods generate either images [25, 17] or masks [19, 7], not both. Simultaneous image-mask
24 generation remains underexplored, especially with flexible, controllable conditioning—critical for
25 simulation, data augmentation, and rare-case modeling.

26 We present **CoSimGen**, a diffusion-based framework for **Controllable Simultaneous image and**
27 **segmentation mask Generation**. CoSimGen supports conditioning on either class labels or natural
28 language prompts and unifies multimodal control in a single generation process (Figure 1). Built
29 upon a Conditional Denoising Diffusion Probabilistic Model (DDPM) with a U-Net backbone [29],
30 CoSimGen introduces two novel components: (1) *Spectron*: a spatio-spectral embedding fusion
31 module that injects class and timestep embeddings into the network. Here, class features are fused
32 spatially to guide object placement and structure; timestep embeddings are fused along channels
33 to model denoising dynamics; and (2) *Textron*: a text-grounded semantic conditioning module that
34 aligns class embeddings with language embeddings, enabling text prompts to be "hot-swapped" in
35 place of class labels during inference. Contrastive learning aligns the embedding spaces to achieve
36 this, as inspired by CLIP [24].

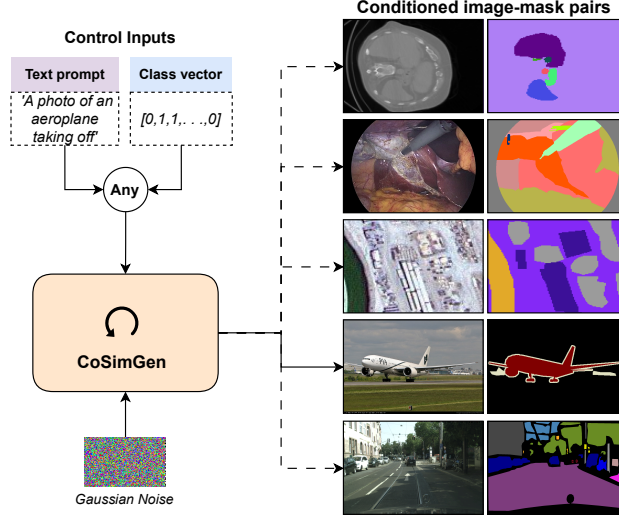


Figure 1: CoSimGen generates paired image and segmentation mask from a class or text prompt. Outputs are high-resolution, semantically aligned, and spatially coherent.

The training objective combines three losses: diffusion loss for denoising, contrastive triplet loss for text-label alignment, and adversarial loss for realism. We first generate low-resolution outputs (128×128) and super-resolve them to (512×512), maintaining fidelity and semantic alignment between image and mask. CoSimGen is evaluated on five diverse datasets—CholecSeg8k [15], BTCV [10], Cityscapes [6], PASCAL VOC [8], and MBRSC [16]—spanning surgical, medical, urban, natural, and satellite imagery domains. We use Fréchet Inception Distance (FID) [13], Kernel Inception Distance (KID) [4], Inception VGG Distance (VGG-D), Learned Perceptual Image Patch Similarity (LPIPS) [34], and Semantic FID (sFID) [3] to assess image quality and semantic realism. For mask-image alignment fidelity, we use sFID and Positive Predictive Value (PPV). CoSimGen outperforms strong baselines across all metrics and datasets. It enables controllable, high-resolution, semantically consistent generation of annotated data, offering a scalable alternative to manual labeling. Our approach is especially valuable in high-stakes domains like surgical training and medical diagnosis, where precise region-level control is essential. Furthermore, CoSimGen can serve as a generative pretraining tool, supporting domain adaptation and low-resource learning setups.

By introducing a general-purpose, controllable framework for paired data generation, CoSimGen addresses fundamental challenges in structured prediction and multimodal generative modeling—core areas of interest in multi-modal generative AI integrating vision and language modeling.

2 Related Work

Image generation from single to multimodal synthesis. Generative modeling has progressed from VAEs [18] and GANs [12] to diffusion models [14, 28], which now dominate high-fidelity image synthesis. Conditional GANs like Pix2Pix [17] and StyleGAN variants introduced structure via paired inputs or style codes. Diffusion-based methods, such as DALL·E [25], Imagen [30], and Surgical Imagen [23], enabled text-to-image synthesis with improved realism, but focus primarily on single-modality outputs.

Segmentation and data synthesis. To overcome annotation bottlenecks in segmentation tasks, works like Text4Seg [19] and SegGen [7] generate segmentation masks from text. Yet, they decouple image and mask generation or lack generalizability. Medical approaches such as HVAE [5] generate paired data but offer limited control and domain scope.

Paired image-mask generation. SimGen [3] and DiffuMask [33] explore joint generation of image and mask, but are either domain-restricted or conditionally limited. OVDiff [21] relies on fixed vocabularies for segmentation post-generation. Prior work lacks a unified, controllable model for simultaneous image-mask synthesis across domains. Our work addresses this gap with fine-grained conditionality, enabling scalable, multimodal generation from text or class vectors.

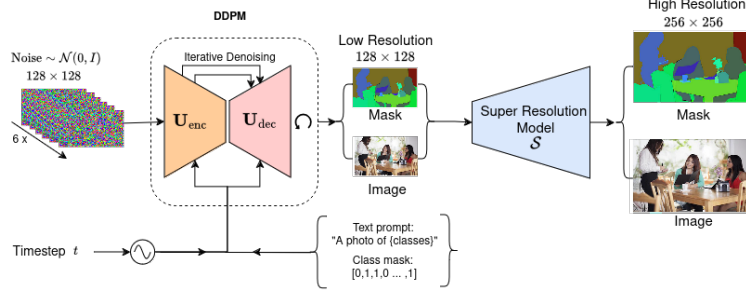


Figure 2: Architecture of CoSimGen for conditional generation of paired image-mask.

3 Methods

Our goal is to generate paired images and semantic segmentation masks, guided by user input prompts, for synthetic data creation and educational purposes. To achieve this, we propose a Controllable Simultaneous Image-Mask Generator (CoSimGen), a diffusion-based framework that utilizes contrastive learning to seamlessly condition generation on text or class labels, ensuring precise alignment between images and masks in a unified process.

3.1 Task Formalization

Let $\mathcal{D} = \{(\mathbf{X}_i, \mathbf{y}_i)\}_{i=1}^n$, where $\mathbf{X}_i \in \mathbb{R}^{C \times H \times W}$ represents an image, and $\mathbf{y}_i \in \{0, 1\}^{H \times W}$ is the corresponding segmentation mask. The mask $\mathbf{y}_i$ contains the segmented objects, which are associated with class labels from $\mathcal{C} = \{c_1, c_2, \dots, c_k\}$. The conditioning vector $\mathbf{c}_i$ is derived from the mask $\mathbf{y}_i$ and encodes the present classes. Alternatively, a text prompt $\mathbf{t}_i$ can be provided as a caption of the image, limited to describing only the objects present in the segmentation mask. The task is to train a model $\mathcal{M}$ that generates image-mask pairs $(\hat{\mathbf{X}}, \hat{\mathbf{y}})$ simultaneously, conditioned on the class labels $\hat{\mathbf{c}}$ and/or the text prompt $\hat{\mathbf{t}}$, such that the generated segmentation mask $\hat{\mathbf{y}}$ aligns with the generated image $\hat{\mathbf{X}}$ and both are similar to real samples from $\mathcal{D}$. The goal is to maximize the likelihood of generating paired data that closely resembles real samples, conditioned on the class labels or the textual description of the segmented objects.

3.2 Model Architecture

The proposed *CoSimGen* is built on a denoising diffusion process that iteratively refines noisy inputs into clean, coherent, paired image-mask samples. As illustrated in Fig. 2, the architecture comprises three main modules: (a) a **low-resolution (LR) generator** that establishes semantic alignment between the image, segmentation mask, and conditioning input prompt, (b) a **conditioning mechanism** that integrates text/class and timestep embeddings to enable flexible, user-controllable generation, and (c) a **super-resolution (SR) module** that upscales the LR coarse outputs into high-resolution (HR) spatial dimensions while preserving alignment and fidelity.

3.2.1 Low-Resolution Image-Mask Generation

CoSimGen employs a U-Net [29] diffusion backbone that leverages a Conditional Denoising Diffusion Probabilistic Model (DDPM) [14]. The U-Net consists of an encoder $\mathbf{U}_{\text{enc}}$ and decoder $\mathbf{U}_{\text{dec}}$ connected via two residual skip connections at each level and trained to denoise a noisy input $\mathbf{X}_t$ over multiple timesteps t . The U-Net is conditioned on semantic (class or text) and timestep embeddings via the mechanisms defined in Sec. 3.2.2, enabling the model to adaptively align its features with both context and noise-level information, guiding the network to produce class-consistent outputs during denoising. This backbone takes as input:

1. A noisy input $\mathbf{X}_t$, representing a corrupted image-mask pair at diffusion step t
2. A binary class mask $\mathbf{M} \in \{0, 1\}^c$, indicating the queried classes
3. A text prompt $\mathbf{Z}_q$, e.g., “A photo of {class}”
4. The timestep value t , representing the current noise level.

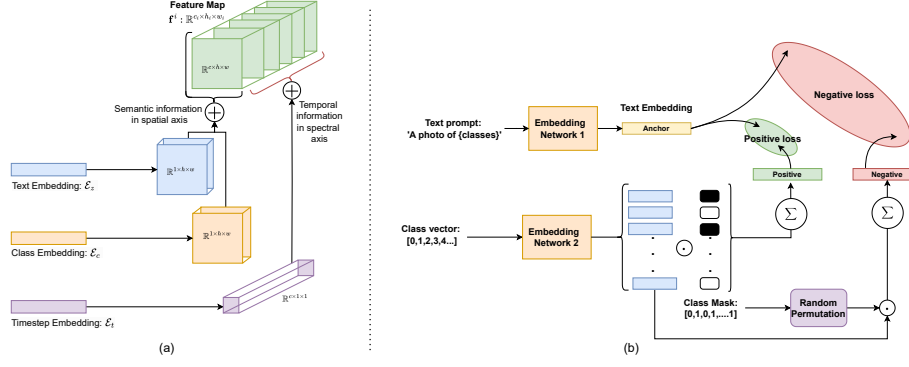


Figure 3: Conditioning mechanisms used in CoSimGen showing: (a) Spectron: spatio-spectral embedding fusion for semantic and temporal conditioning, and (b) Textron: contrastive alignment for interchangeable class/text conditioning.

3.2.2 Conditioning Mechanisms

Text encoder. The text encoder $\mathcal{E}_z$ processes input text $\mathbf{Z}_q$ using a frozen sentence transformer [27], followed by projection layers that embed it into a D -dimensional space: $\mathbf{Z}_{\text{emb}} \in \mathbb{R}^{1 \times D} = \mathcal{E}_z(\mathbf{Z}_q, \theta_T)$

Class encoder. Given a binary class mask $\mathbf{M} \in \{0, 1\}^c$, the class encoder $\mathcal{E}_c$ multiplies $\mathbf{M}$ with a learnable weight matrix $\mathbf{W}_c \in \mathbb{R}^{c \times d}$ effectively selecting the active class embeddings and sums across classes: $\mathbf{C}_{\text{feat}} = \sum_{i=1}^c \mathbf{M}_i \cdot \mathbf{W}_c[i, :]$. The $\mathbf{C}_{\text{feat}}$ passes through a series of linear transformations projecting it into a D -dimensional class embedding: $\mathbf{C}_{\text{emb}} \in \mathbb{R}^{1 \times D} = \mathcal{E}_c(\mathbf{M}; \theta_{\mathcal{E}_c})$

Timestep encoder. The timestep encoder $\mathcal{E}_t$ maps scalar timestep t into a D -dimensional embedding via sinusoidal positional encoding followed by MLP projections: $\mathbf{T}_{\text{emb}} \in \mathbb{R}^{1 \times D} = \mathcal{E}_t(t, \theta_{\mathcal{E}_t})$

These conditional embeddings ($\mathbf{Z}_{\text{emb}}, \mathbf{C}_{\text{emb}}, \mathbf{T}_{\text{emb}}$) are intuitively injected into the model via two approaches which we proposed in this work:

(a) **Spatio-spectral embedding fusion (Spectron):** In traditional generative models, conditioning feedback is applied by direct concatenation along the latent space or by adding conditions to feature maps along the channel axis. Diffusion models often follow a similar approach, where the conditional embeddings and timestep embeddings are introduced by adding them to the channel axis of the feature maps. While effective, this approach does not fully exploit the semantic richness of the conditional embeddings.

To bridge this gap, we introduce *Spectron* (Fig. 3a), a strategy that injects conditions into feature maps at all resolutions, allowing for a more intuitive conditioning process. Recognizing that class conditions, such as the class embedding $\mathbf{C}_{\text{emb}}$, represent a semantic understanding of the image and mask, we propose spatially embedding this information. This semantic representation governs the shape, outline, and textures within the generated image and mask. Therefore, it is intuitively powerful to apply the semantic conditional vectors along the spatial dimensions, thereby spatially conditioning the features $\mathbf{f}$ at each resolution i :

$$\mathbf{f}_{\text{cond}}^{i, \text{spatial}} = \mathbf{f}^i + \mathbf{C}_{\text{emb}}^i \quad (1)$$

where $\mathbf{f}^i : \mathbb{R}^{c_i \times h_i \times w_i}$ and $\mathbf{C}_{\text{emb}}^i : \mathbb{R}^{1 \times h_i \times w_i}$ thus adding the conditional embedding in the spatial dimension.

The timestep embedding $\mathbf{T}_{\text{emb}}$, by contrast, encodes the noisiness of the input, and the noise level is assumed to affect all channels uniformly and equally. Hence, it becomes intuitive to apply the timestep conditioning along the channel dimension of the noisy feature maps, thereby spectrally conditioning the features $\mathbf{f}$ at each resolution i :

$$\mathbf{f}_{\text{cond}}^{i, \text{spectral}} = \mathbf{f}_{\text{cond}}^{i, \text{spatial}} + \mathbf{T}_{\text{emb}}^i \quad (2)$$

where $\mathbf{f}_{\text{enc}}^{i, \text{spectral}} : \mathbb{R}^{c_i \times h_i \times w_i}$ and $\mathbf{T}_{\text{emb}}^i : \mathbb{R}^{c_i \times 1 \times 1}$. By combining these two perspectives, Spatio-Spectral Feature Mixing enables both semantic and temporal feedback, allowing $\mathbf{U}$ to generate

features that are spatially aligned with the condition semantics and spectrally aligned with the temporal noise level. This dual conditioning mechanism ensures that the U-Net captures a deep alignment between the class condition and timestep information across all spatial and spectral dimensions, enhancing the model’s generative capabilities.

- (b) **Text-grounded class conditioning (Textron):** While class embeddings $\mathbf{C}_{\text{emb}}$ can condition image and mask generation independently, they lack the flexibility to allow inference on text inputs directly. To address this, *Textron* aligns class embeddings with their corresponding text embeddings during training, enabling the model to accept either class or text embeddings interchangeably during inference. This is achieved by learning a shared embedding space where class and text representations are closely aligned.

Conventional generative models condition on text by learning a similarity metric between text and generated image features. While effective for evaluating alignment, this approach does not enable *direct substitution* (or “hot-swapping”) of class embeddings with text embeddings during inference. Textron overcomes this limitation by introducing a contrastive triplet loss (Eq. 3) that explicitly aligns class and text embeddings. Given a text embedding $\mathbf{Z}_{\text{emb}}$ (anchor), the corresponding class embedding $\mathbf{C}_{\text{emb}}$ (positive), and a mismatched class embedding $\tilde{\mathbf{C}}_{\text{emb}}$ (negative), the loss is defined as:

$$\mathcal{L}_{\text{triplet}} = \max \left(0, \|\mathbf{Z}_{\text{emb}} - \mathbf{C}_{\text{emb}}\|^2 - \|\mathbf{Z}_{\text{emb}} - \tilde{\mathbf{C}}_{\text{emb}}\|^2 + \alpha \right) \quad (3)$$

This objective encourages the model to reduce the distance between matched text-class pairs while pushing apart mismatched ones, with margin α controlling the separation. As illustrated in Fig. 3(b), the text embedding $\mathbf{Z}_{\text{emb}}$ serves as the anchor; the corresponding class embedding $\mathbf{C}_{\text{emb}}$ is used as the positive, and a randomly selected class embedding $\tilde{\mathbf{C}}_{\text{emb}}$ forms the negative. By minimizing this loss, the model learns a unified embedding space where class and text representations are close, enabling the class encoder to be replaced with a text encoder during inference. This design allows the model to leverage the semantic richness of natural language, supporting flexible and efficient text-grounded generation.

3.2.3 Super-Resolution Module

Upscaling strategy. To enhance visual fidelity, the coarse outputs $\mathbf{X}_{\text{LR}} \in \mathbb{R}^{6 \times 128 \times 128}$ are passed to a super-resolution model ($\mathcal{SR}$). We adopt an Efficient Sub-Pixel CNN (ESPCNN) [31] as the $\mathcal{SR}$ with an upscale factor of 2.

Training objective. During training, Gaussian noise is added to the ground-truth low-resolution input:

$$\tilde{\mathbf{X}}_{\text{LR}}^{\text{gt}} = \mathbf{X}_{\text{LR}}^{\text{gt}} + \epsilon, \quad \mathcal{SR}(\tilde{\mathbf{X}}_{\text{LR}}^{\text{gt}}) \rightarrow \mathbf{X}_{\text{HR}}^{\text{gt}} \in \mathbb{R}^{6 \times 2h \times 2w}$$

The SR model is trained progressively from $128 \times 128 \rightarrow 256 \times 256 \rightarrow 512 \times 512$, supporting multi-scale inference.

4 Experiments

Datasets. We evaluate on five diverse segmentation datasets: Cityscapes [6], PASCAL VOC [8], MBRSC [16], BTCV [10], and CholecSeg8k [15], covering general, remote sensing, radiology, and surgical domains. To enhance class separability, segmentation masks are mapped to a uniformly spaced Fibonacci RGB (F-RGB) space using a golden angle transformation.

Implementation. CoSimGen is trained at 128×128 resolution, with a residual U-Net backbone ($d = 64$, multipliers $1 : 8$). A separate super-resolution module ($\mathcal{SR}$) scales outputs to 256^2 and 512^2 . We use Adam optimizer ($\text{lr} = 2 \times 10^{-4}$, batch size 24), PyTorch mixed-precision training, and NVIDIA H100 GPUs running for under 72 training hours.

Loss Functions. CoSimGen is optimized using a combination of noise prediction, alignment, adversarial, and perceptual objectives. The core diffusion model minimizes a conditional noise reconstruction loss $\mathcal{L}_{\text{diff}}$, guided by timestep t , text embedding $\mathbf{T}_{\text{emb}}$, and class embedding $\mathbf{C}_{\text{emb}}$. To enforce semantic alignment, we introduce a triplet loss $\mathcal{L}_{\text{triplet}}$ that attracts $\mathbf{T}_{\text{emb}}$ to $\mathbf{C}_{\text{emb}}$ and repels it from a negative class embedding. Realism of generated image-mask pairs is encouraged via an adversarial loss $\mathcal{L}_{\text{adv}}$ against a frozen discriminator. For high-resolution refinement, a super-resolution loss $\mathcal{L}_{\text{SR}}$ minimizes a weighted combination of MSE and perceptual loss. The full objective

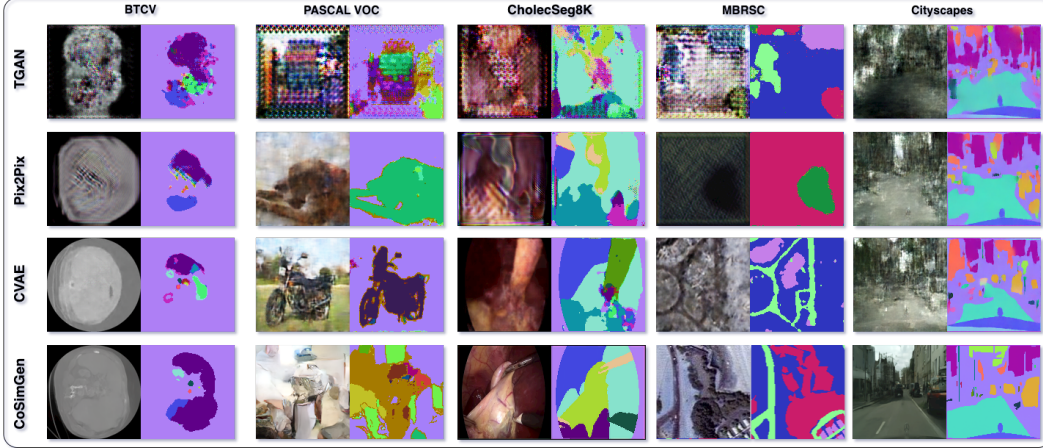


Figure 4: Qualitative comparisons of generated image-mask pairs (low-resolution outputs)

is: $\mathcal{L}_{\text{CoSimGen}} = \mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{triplet}} + \beta \cdot \mathcal{L}_{\text{adv}} + \mathcal{L}_{\text{SR}}$. We provide detailed formulations for each loss component in Appendix.

Baselines. Given the novel challenge of entangled generation of image-mask pairs, we evaluate both regression-based and adversarial generative approaches. For adversarial methods, we adapt classical frameworks, including TGAN [26] and Pix2PixGAN [17], to handle dual outputs. As a regression-based baseline, we employ a conditional convolutional Variational Autoencoder (CVAE) [18] with joint image-mask reconstruction. All baselines are tuned for fairness across datasets. While recent generative models have shown strong results in single-modality settings (e.g., image-only or segmentation-only), directly applying them to the coupled image-mask generation task proves non-trivial. We observed that such models require extensive architectural modifications to jointly handle continuous and discrete modalities—often reducing them to their foundational counterparts. Thus, our baseline comparisons focus on principled, extensible variants of standard models, reflecting a fair and interpretable benchmark for CoSimGen.

Evaluation Protocols. We report Fréchet Inception Distance (FID) [13], Kernel Inception Distance (KID) [4], Learned Perceptual Image Patch Similarity (LPIPS) [34], and Inception VGG Distance (VGG-D) between real and generated images to assess visual realism. To evaluate mask-image alignment and regional generation fidelity, we compute *semantic FID* (sFID) [3], which extend the traditional FID to class-specific image regions guided by generated masks. These metrics quantify how well generated regions match target semantics. Further details on sFID are provided in Appendix. Conditioning quality is measured via Positive Predictive Value (PPV) of queried classes in the generated masks. For the super-resolution module $\mathcal{SR}$, we evaluate reconstruction quality by training on generated low- and high-resolution pairs and testing on low-resolution inputs.

5 Results

5.1 Image Quality

We evaluate the quality of generated image-mask pairs both qualitatively and quantitatively. Fig. 4 provides visual comparisons between CoSimGen and baseline models — CVAE, TGAN, and Pix2PixGAN — across diverse datasets. CoSimGen consistently produces sharper, crisper, more coherent images with structurally aligned masks, particularly excelling in datasets with ample training data. This fidelity is especially evident in complex domains like surgical scenes and urban layouts, where both visual detail and semantic alignment are critical.

While CVAE performs competitively on smaller datasets such as PASCAL VOC, reflecting its advantage in low-data regimes, CoSimGen significantly outperforms all baselines on larger-scale datasets, including Cityscapes, CholecSeg8k, MBRSC, and BTCV. This performance gap underscores the scalability of CoSimGen, driven by its design to handle high-resolution structures and complex spatial distributions.

Table 1: Evaluation of the fidelity of the generated images across 3 datasets in comparison with the baselines across four metrics: FID, KID, VGG distance, and LPIPS distance.

Model	Pascal VOC				MBRSC				BTCV			
	FID	KID	VGG-D	LPIPS-D	FID	KID	VGG-D	LPIPS-D	FID	KID	VGG-D	LPIPS-D
TGAN	348.19	0.29	221.53	0.77	394.86	0.27	113.09	0.72	394.05	0.53	146.31	0.60
Pix2PixGAN	348.05	0.30	225.56	0.79	410.18	0.34	117.94	0.70	284.60	0.36	152.80	0.54
CVAE	337.41	0.35	204.97	0.76	326.16	0.27	106.79	0.70	192.21	0.19	144.84	0.45
CoSimGen (Ours)	206.29	0.20	227.64	0.74	203.67	0.11	110.43	0.63	159.92	0.13	139.35	0.53

Table 2: Comparative evaluation of the generated mask-image alignment in 4 datasets across two metrics: semantic fr chet inception distance (sFID) and positive predicted value (PPV).

Model	Cityscapes		Pascal VOC		MBRSC		BTCV	
	sFID	PPV	sFID	PPV	sFID	PPV	sFID	PPV
TGAN	345.29±39.21	0.68±0.08	348.21±54.32	0.98±0.02	435.12±20.87	0.43±0.02	405.07±23.52	0.50±0.09
Pix2PixGAN	128.12±12.09	0.92±0.03	326.66±87.23	0.81±0.06	462.74±19.15	0.84±0.06	323.26±28.25	0.43±0.04
CVAE	204.27±9.82	0.86±0.10	381.53±29.11	0.90±0.07	422.80±33.10	0.91±0.08	250.52±17.67	0.56±0.09
CoSimGen (Ours)	54.08±8.93	0.98±0.01	343.66±10.89	0.78±0.01	294.66±12.20	0.87±0.00	198.74±5.68	0.35±0.00

The visual analysis in Fig. 4 reveals key qualitative distinctions. TGAN and Pix2PixGAN, while visually plausible in isolated cases, suffer from mode collapse, texture artifacts, and misaligned masks. In contrast, CoSimGen preserves semantic boundaries with high fidelity, generating clinically plausible, spatially consistent outputs. Compared to CVAE, CoSimGen offers crisper structural detail and greater diversity in textures—especially visible in Cityscapes and surgical datasets.

Quantitative results in Table 1 reinforce these findings. CoSimGen achieves the lowest FID, KID, and LPIPS scores on four of the five datasets (MBRSC, BTCV, CholecSeg8k, Cityscapes), demonstrating superior realism and perceptual quality. Notably, our model sets a new benchmark on BTCV and MBRSC in both image quality and semantic fidelity. For radiology datasets, where accurate anatomical delineation is essential, CoSimGen maintains high mask-image coherence—outperforming adversarial models and matching or exceeding CVAE.

Overall, CoSimGen delivers state-of-the-art visual and semantic quality across a wide range of image domains, capturing both global realism and fine-grained structural alignment. Additional samples and visual comparisons are provided in the Appendix.

5.2 Image-Mask Alignment

Table 2 reports the alignment quality between generated images and segmentation masks across five datasets, using Semantic FID (sFID) and Positive Predictive Value (PPV). CoSimGen achieves the best sFID scores on Cityscapes, MBRSC, and BTCV, confirming its strong ability to preserve semantic consistency at the regional level. It also ranks second in PPV on MBRSC and BTCV, demonstrating reliable class-conditional generation. Performance on CholecSeg8k mirrors that on Cityscapes, reaffirming CoSimGen’s alignment strength even in surgical domains with complex spatial priors.

On Pascal VOC, CoSimGen lags behind, which we attribute to the dataset’s small size and high variance. Interestingly, CVAE performs better in this low-resource setting, suggesting VAE-style reconstruction is more stable when semantic structure is weakly represented. However, TGAN’s deceptively high PPV stems from repetitive outputs—highlighting a trade-off between mask correctness and sample diversity, which CoSimGen manages more effectively.

Across datasets, results indicate that CoSimGen scales better with data complexity and maintains high semantic alignment under diverse conditions.

5.3 Input-Output Alignment

We assess alignment between input prompts and generated outputs using semantic FID (sFID), which measures class-wise fidelity by comparing semantic regions of generated images to real ones. Lower scores indicate both accurate class presence and precise spatial consistency. As reported in Table 2, CoSimGen achieves substantially lower sFID across all datasets, confirming reliable grounding of

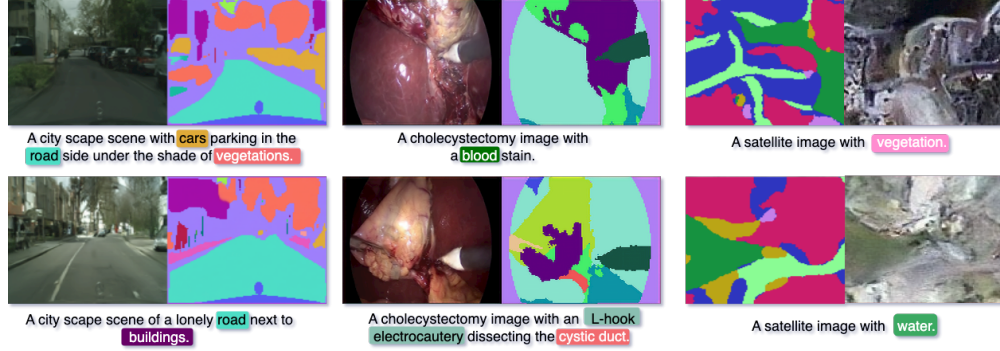


Figure 5: Qualitative results showing text(class)-conditioned image-mask generation

generation in the prompted classes. Visualizations in Fig. 5 further support this: masks not only include all requested classes, but their spatial layout reflects scene plausibility, which is also preserved in the corresponding images. Unlike baselines, which often miss prompted classes or distort their geometry, CoSimGen maintains both semantic and structural consistency. These results underscore the model’s capacity to respect discrete mask structure while simultaneously generating continuous images—an essential requirement for controllable, high-quality surgical data synthesis.

5.4 High-Resolution (HR) Outputs

The super-resolution (SR) images produced by ESPCNN [31], utilized in our proposed CoSimGen framework, are compared with baseline outputs from SRGAN [20] on CholecSeg8K [15] and BTCV [10] datasets. The results in Fig. 6 demonstrate that ESPCNN effectively captures high-frequency details that SRGAN fails to reproduce. This distinction is particularly evident in the sharper boundaries between textures, such as those of organs, bones, and blood vessels, highlighting ESPCNN’s superior ability to preserve structural details. More results are provided in the Appendix.

Detailed ablation results analyzing the impact of CoSimGen’s core contributions, triplet loss for text-grounded alignment and discriminator loss for fidelity regularization, along with their combined effect, are provided in the Appendix to demonstrate their individual and complementary benefits.

5.5 Discussion

Our experiments reveal that models optimized with regression objectives exhibit greater stability compared to those with adversarial objectives, particularly in tasks involving joint estimation of continuous and discrete distribution pairs. In practice, we observed that adversarial models frequently suffered from mode collapse, which undermines their reliability for such tasks. By contrast, regression-based approaches minimize prediction error in a smoother optimization landscape, which contributes to improved stability. We also found that incorporating adversarial loss as a regularizer in our model introduced oscillatory behavior in early training stages, where generation quality fluctuates. However, as training progresses, these oscillations diminishes, and generation quality stabilizes. This suggests that adversarial loss, while beneficial as a regularizer, may require careful tuning to balance stability with generation fidelity.

5.6 Limitations

CoSimGen, like most diffusion-based generative models, is highly data-dependent. It requires substantial amounts of annotated segmentation masks paired with class-specific text labels to perform effectively. Our results show a noticeable drop in performance on datasets with limited samples, such as PASCAL VOC, indicating that CoSimGen is optimized for high-fidelity generation in moderate to large-scale data settings. While data augmentation techniques (e.g., random rotations and flips) can partially mitigate this limitation in specific domains like satellite imagery, their applicability to natural or medical scenes is limited. This highlights a key challenge: the need for more effective augmentation strategies or lightweight adaptations of CoSimGen to make it viable for low-resource and few-shot scenarios.

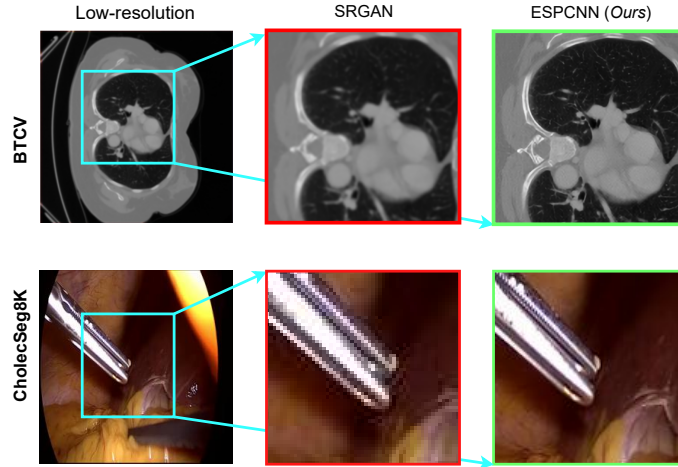


Figure 6: Comparison of super-resolution result of ESPCNN (in our model) and SRGAN (baseline).

Moreover, diffusion training is computationally intensive, which may restrict accessibility to users without high-end compute resources. Developing more efficient variants of CoSimGen or integrating it with faster approximation techniques could broaden its usability.

In terms of scope, all datasets used in this study lack identifiable human features. As such, we were unable to evaluate the model’s behavior on data involving personally identifiable information. Users should exercise caution when deploying CoSimGen in privacy-sensitive settings, as its privacy-preserving capabilities remain unverified.

Although CoSimGen demonstrates strong input-output alignment and fidelity on curated benchmarks, its generalizability to out-of-distribution prompts or unseen object categories has not been systematically evaluated, leaving open questions about robustness in more diverse real-world applications.

6 Conclusion

This work introduces CoSimGen, a novel diffusion-based framework for controllable simultaneous image and segmentation mask generation. By addressing the critical challenges in existing generative models, CoSimGen provides a unified solution for producing high-quality paired datasets with precise control during generation. The model leverages text-grounded class conditioning, spatial-temporal embedding fusion, and multi-loss optimization, enabling robust performance across applications requiring spatial accuracy and flexibility. CoSimGen demonstrates state-of-the-art performance on diverse datasets, making it a versatile tool for augmenting datasets, simulating rare scenarios, and tackling domain-specific challenges. Its outputs offer a scalable alternative to manual annotation, significantly reducing the time and resources required for dataset creation. Moreover, the generated paired data serve as a ready source for pretraining models, given the framework’s ability to produce an unlimited variety of high-fidelity, condition-adherent examples. Beyond its utility in dataset augmentation, CoSimGen establishes a foundation for future research in multi-modal, multi-class, and domain-adaptive generative frameworks. By bridging the gap between generative AI and real-world applications, the framework addresses critical bottlenecks in precision-driven and privacy-sensitive domains, advancing cross-domain AI research and deployment. CoSimGen represents a significant step forward in enabling scalable, controllable data generation, unlocking new possibilities for pretraining, robustness testing, and real-world impact.

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